The Effect of an AI-Assisted Research Workshop on Early-Career Earnings: Evidence from a Waitlist Design

Workshop Participant

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Abstract

This paper evaluates the effect of a five-session workshop on AI-assisted academic research using Claude Code. We compare 150 attendees to 80 individuals on the waitlist using a simple pre–post design. Attendees earned approximately $16,384 more than waitlisted individuals two years after the course, controlling for age and prior experience. Satisfaction scores increased across sessions, though cumulative attrition reached 37% by course completion. All data are synthetic and generated for pedagogical purposes.

1 Introduction

Can a short workshop on AI-assisted research tools meaningfully affect researchers' productivity and earnings? This paper examines that question using data from a five-session course on Claude Code, an AI coding assistant designed for academic research workflows. We exploit a natural comparison group — individuals on the course waitlist — to estimate the effect of attendance on post-course earnings.

The workshop covered data cleaning, estimation, robustness checks, LaTeX integration, version control, and advanced features such as hooks, MCP servers, and custom skills. Participants ranged from doctoral students to mid-career researchers across economics, statistics, and computational fields.

2 Data

Our sample consists of 230 individuals: 150 who attended the workshop and 80 who were placed on a waitlist. Table 1 presents summary statistics for both groups.

Table 1: Summary Statistics by Attendance Status

Variable	Attended (Mean)	Attended (SD)	Waitlist (Mean)	Waitlist (SD)
Age	34.34	7.54	36.18	7.99
Experience Years	9.96	6.03	10.84	6.26
N	150		80	

The two groups are broadly comparable at baseline. Attendees are slightly younger on average (34.3 vs. 36.2 years) and have marginally less professional experience (10.0 vs. 10.8 years). These differences are small relative to the within-group standard deviations, suggesting that waitlist assignment was approximately random with respect to observable characteristics.

We observe pre-course and post-course earnings (measured two years after the workshop) for all 230 individuals. Satisfaction scores, collected at the end of each session, are available only for attendees.

3 Course Enrollment and Attrition

Figure 1 shows the number of students remaining at each session. Of the 150 who began the course, 94 completed all five sessions — a cumulative attrition rate of 37%.

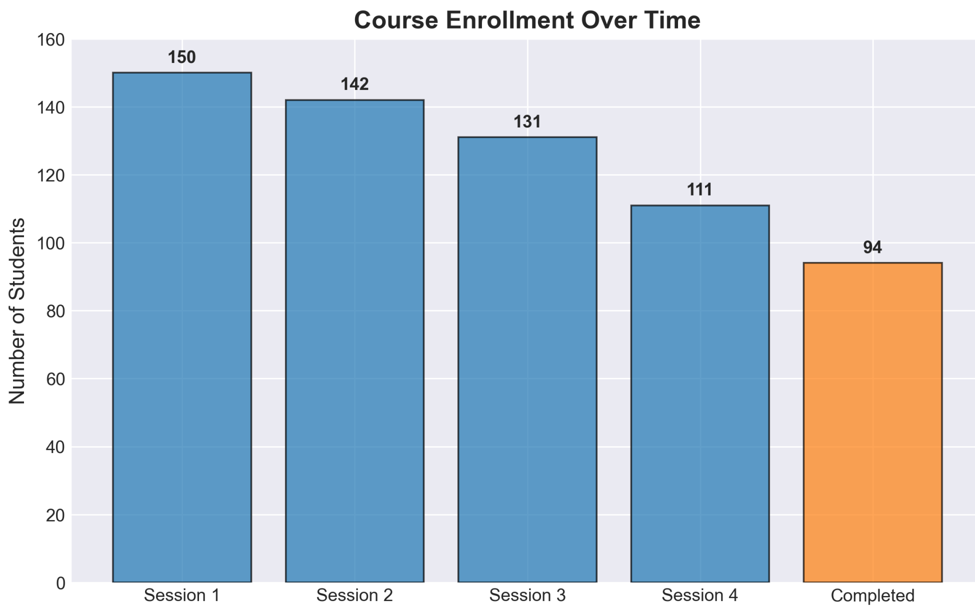

Figure 1: Course enrollment over time. The final bar indicates the number of students who completed all sessions.

Figure 2 breaks this down by transition. Attrition accelerated over the course: 5% between Sessions 1 and 2, rising to 15% between Session 4 and course completion. This pattern is consistent with increasing technical difficulty in later sessions (hooks, MCP configuration, skills), though we cannot rule out scheduling conflicts or selection effects.

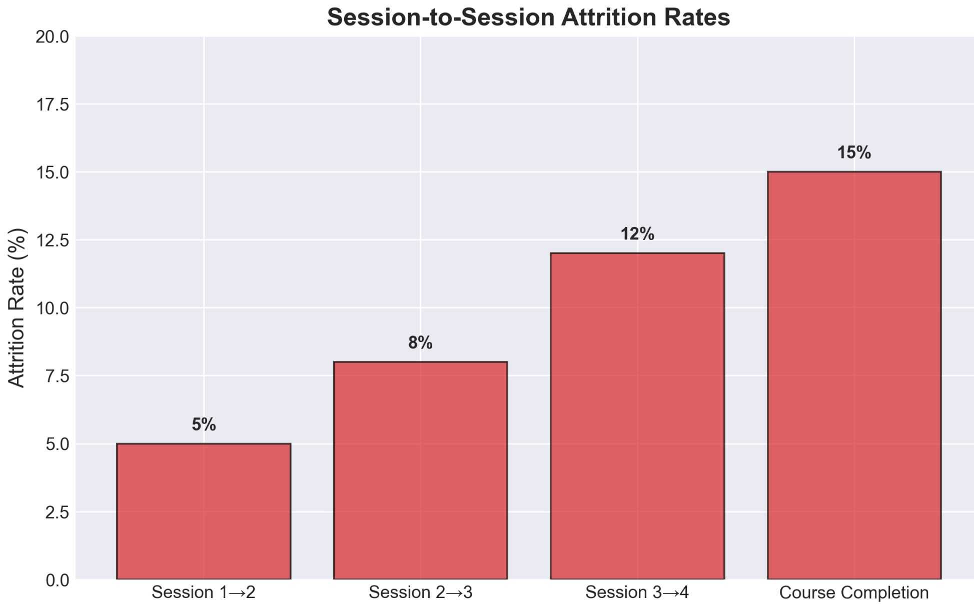

Figure 2: Session-to-session attrition rates.

4 Student Satisfaction

Figure 3 plots mean satisfaction scores across sessions, with the shaded region indicating one standard deviation. Mean satisfaction rose from 7.5 in Session 1 to 8.7 in the final evaluation, a steady increase that suggests the course built value as it progressed rather than front-loading engagement.

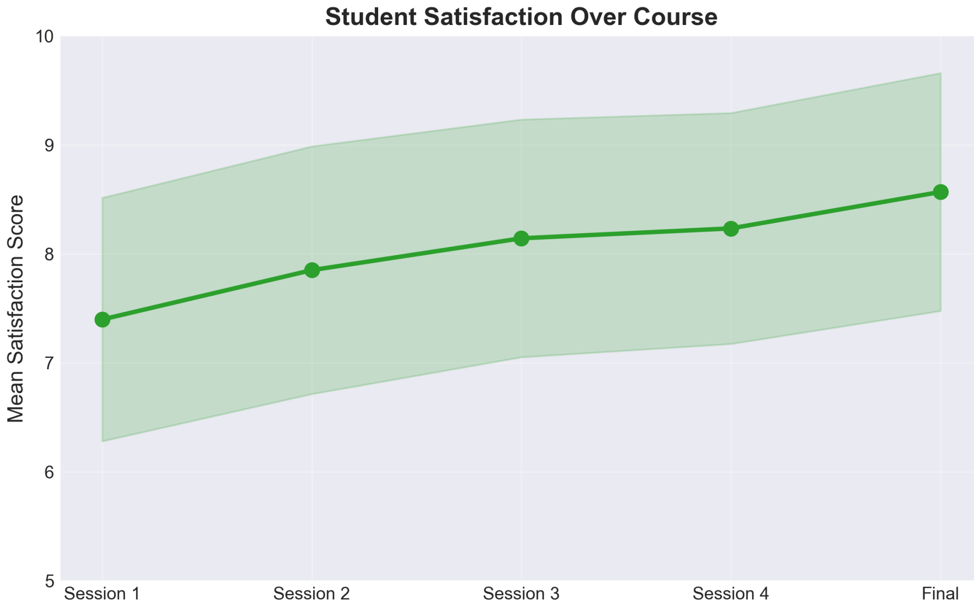

Figure 3: Mean satisfaction score across sessions. Shaded region indicates ± 1 standard deviation.

A caveat: satisfaction scores are observed only for students who remained enrolled. If dissatisfied students were more likely to drop out, the upward trend partly reflects survivorship bias rather than genuine improvement in perceived course quality.

5 Results

5.1 Main Specification

We estimate the effect of course attendance on post-course earnings using OLS:

$$\text{Earnings}_i = \beta_0 + \beta_1 \text{Attended}_i + \beta_2 \text{Experience}_i + \beta_3 \text{Age}_i + \varepsilon_i \tag{1}$$

Table 2 reports the results.

Table 2: OLS Regression: Effect of Course Attendance on Post-Course Earnings

Variable	Coefficient	(Std. Error)
Attended Course	16384.42***	(3093.71)
Experience (years)	-269.67	(251.17)
Age	310.03**	(97.79)
Constant	73391.16	
Observations	230	
R^2	0.1046	

Note: Dependent variable is post-course earnings in USD. Standard errors in parentheses.

Significance levels: $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

Course attendance is associated with an increase in post-course earnings of approximately $16,384, significant at the 0.1% level. Age is positively associated with earnings ($310 per year, $p < 0.01$), while years of experience shows no statistically significant relationship in this specification. The model explains roughly 10% of the variation in post-course earnings.

5.2 Earnings Distribution

Figure 4 compares the distribution of post-course earnings between the two groups. The attended group shows both a higher median and a higher mean (red diamonds), consistent with the regression estimate.

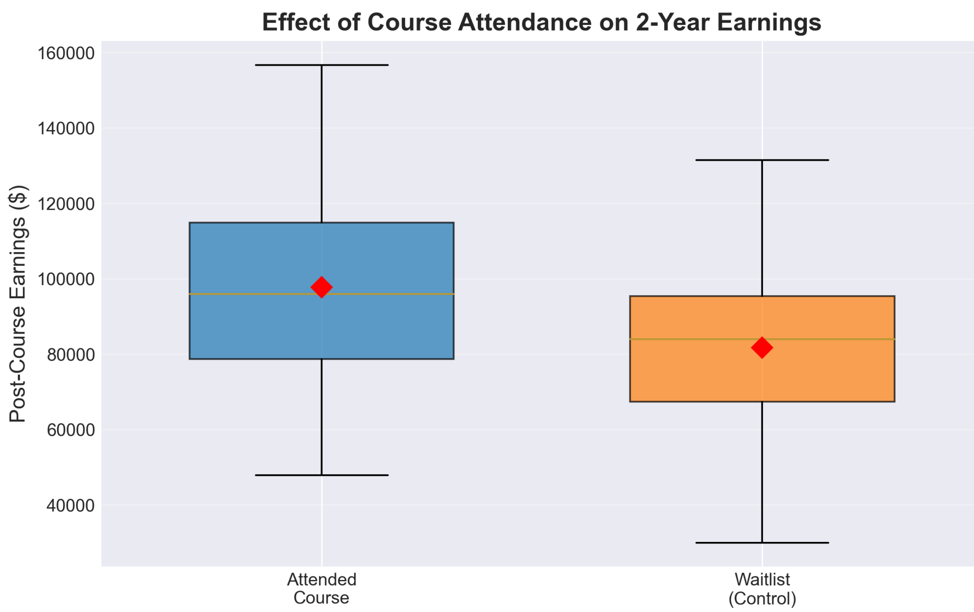

Figure 4: Post-course earnings by attendance status. Red diamonds indicate group means.

6 Discussion

The estimated earnings effect of $16,384 is large relative to the cost of a five-session workshop. Several caveats are in order. First, the waitlist comparison group is not a true randomized control: unobserved motivation or institutional support may differ between those who secured a spot and those who did not. Second, the R^2 of 0.10 indicates that most earnings variation is unexplained by our model, leaving room for omitted variable bias. Third, all data in this analysis are synthetic — the results are generated for pedagogical purposes and do not reflect any real intervention.

7 Conclusion

This paper demonstrates a complete research workflow — from data generation through summary statistics, regression analysis, and figure production — using Claude Code as an AI-assisted research tool. The exercise illustrates how a LaTeX document can be integrated into a version-controlled project directory alongside the code and data that produce its tables and figures.